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import java.io.*;
import java.util.ArrayList;

public class GradeAnalyzer {
    private static int invalidLinesSkipped;

    public static void main(String[] args) {
        ArrayList<Integer> scores = readScores("scores.txt");

        if (scores.isEmpty()) {
            System.out.println("No valid scores were found. Report not created."
);
            return;
        }

        int high = Integer.MIN_VALUE;
        int low = Integer.MAX_VALUE;

        for (int score : scores) {
            if (score > high) {
                high = score;
            }
            if (score < low) {
                low = score;
            }
        }

        double average = calculateAverage(scores);
        writeReport(scores, average, high, low, "report.txt");
    }

    // Returns a list of valid scores read from the file
    public static ArrayList<Integer> readScores(String filename) {
        ArrayList<Integer> scores = new ArrayList<>();
        invalidLinesSkipped = 0;

        try (BufferedReader reader = new BufferedReader(new FileReader(filename)
)) {
            String line;

            while ((line = reader.readLine()) != null) {
                line = line.trim();

                if (line.isEmpty()) {
                    invalidLinesSkipped++;
                    continue;
                }

                try {
                    scores.add(Integer.parseInt(line));
                } catch (NumberFormatException exception) {
                    System.out.println("Skipping invalid score: " + line);
                    invalidLinesSkipped++;
                }
            }
        } catch (IOException exception) {
            System.out.println("Could not read " + filename + ": " + exception.g
etMessage());
        }

        return scores;
    }
}

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// Returns the average of a list of scores, or 0.0 if the list is empty
public static double calculateAverage(ArrayList<Integer> scores) {
    if (scores.isEmpty()) {
        return 0.0;
    }

    double total = 0.0;
    for (int score : scores) {
        total += score;
    }

    return total / scores.size();
}

// Writes and prints the report
public static void writeReport(ArrayList<Integer> scores,
                                double avg, int high, int low,
                                String outputFile) {

    int countA = 0;
    int countB = 0;
    int countC = 0;
    int countD = 0;
    int countF = 0;

    for (int score : scores) {
        if (score >= 90) {
            countA++;
        } else if (score >= 80) {
            countB++;
        } else if (score >= 70) {
            countC++;
        } else if (score >= 60) {
            countD++;
        } else {
            countF++;
        }
    }

    String[] reportLines = {
        "=== Grade Analysis Report ===",
        String.format("Total scores processed:  %d", scores.size()),
        String.format("Invalid lines skipped:    %d", invalidLinesSkipped),
        "",
        String.format("Average score: %.2f", avg),
        String.format("Highest score: %d", high),
        String.format("Lowest score:  %d", low),
        "",
        "Grade distribution:",
        String.format("  A (90-100):    %d", countA),
        String.format("  B (80-89):    %d", countB),
        String.format("  C (70-79):    %d", countC),
        String.format("  D (60-69):    %d", countD),
        String.format("  F (below 60): %d", countF)
    };

    try (BufferedWriter writer = new BufferedWriter(new FileWriter(outputFil
e))) {
        for (String reportLine : reportLines) {
            System.out.println(reportLine);
            writer.write(reportLine);
            writer.newLine();
        }
    }
}

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        }  
    } catch (IOException exception) {  
        System.out.println("Could not write " + outputFile + ": " + exception  
n.getMessage());  
    }  
}
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